

MSU-Bench: Towards Speaker-Centric Understanding in Conversational Multi-Speaker Scenarios

Zhaokai Sun^{1,3,4,*}, Shuai Wang^{2,3,*}, Zhennan Lin^{1,*}, Chengyou Wang¹, Dehui Gao¹, Yuang Cao¹, Chunjiang He¹, Pan Zhou⁴, Lei Xie^{1,**}

¹ Audio, Speech and Language Processing Group (ASLP@NPU),
School of Software, Northwestern Polytechnical University, China

² School of Intelligent Science and Technology, Nanjing University, China

³ Shenzhen Loop Area Institute, China

⁴ Base Model, Li Auto, China

zksun@mail.nwpu.edu.cn, shuaiwang@nju.edu.cn, lxie@nwpu.edu.cn

Abstract

Spoken Language Understanding (SLU) is moving from task-specific pipelines toward large audio language models (LALMs) that generate natural-language responses. However, existing speech benchmarks mainly focus on single-speaker settings or isolated subtasks, leaving speaker-centric understanding in realistic multi-speaker conversations insufficiently evaluated. We introduce MSU-Bench, a diagnostic benchmark for multi-speaker conversational understanding, covering 16 speaker-centric tasks and 2,300 QA instances in a two-tier framework from speaker grounding to dialogue reasoning. We build a Gemini-assisted annotation and QA generation pipeline with human-in-the-loop verification, achieving high QA validity and strong agreement between human answers and verified labels. We further analyze speaker-referencing schemes and diagnostic error types to reveal bottlenecks in speaker grounding and reasoning. Experiments reveal clear gaps across model families, with closed-source systems leading overall but all models still facing challenges in complex speaker grounding and multi-speaker reasoning. The benchmark annotations, metadata, and evaluation scripts will be available at the GitHub repository: [ASLP-lab/MSU-Bench](https://github.com/ASLP-lab/MSU-Bench).

Index Terms: multi-speaker conversational understanding, large audio language models, speaker-centric evaluation

1. Introduction

Spoken language understanding (SLU) aims to interpret speech beyond verbatim transcription, requiring models to jointly capture linguistic content as well as paralinguistic and pragmatic cues. With the emergence of large audio language models (LALMs) [1, 2, 3], SLU is shifting from task-specific pipelines, such as ASR and speaker analysis, to end-to-end audio-to-text generation that unifies perception and reasoning in a single model [4, 5, 6]. However, real-world speech interactions are often conversational and multi-speaker, involving rapid turn switching, interruptions, overlaps, and speaker-dependent variations in speaking rate, emotion, and style. In such settings, conversational understanding cannot rely on acoustic perception or transcript content alone; it also requires models to identify speaker identities, track speaker consistency across turns, understand dialogue structure and interaction relations, and reason over context across turns and speakers.

*These authors contributed equally.

**indicates the corresponding author.

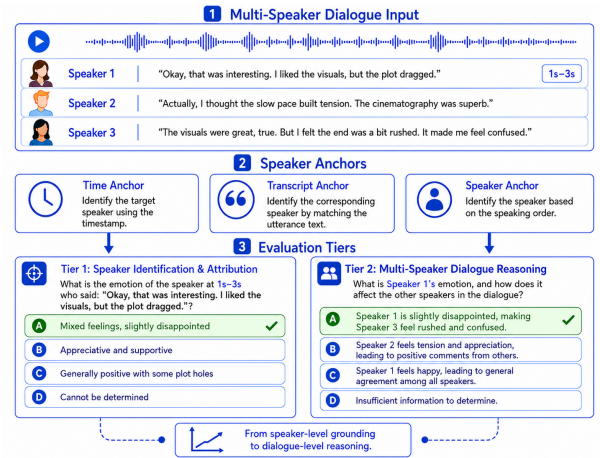


Figure 1: *Two-tier task hierarchy of MSU-Bench. Tasks progress from speaker grounding to multi-speaker reasoning.*

Recent work on speaker understanding in LALMs has begun to incorporate speaker and temporal structure directly into the model rather than relying on external post-processing. Typical approaches generate speaker-attributed transcripts in structured formats, introduce speaker registration for controllable attribution, and represent timestamps with compact time tokens or time-aware encodings [7, 8, 9]. Newer methods further disentangle content from speaker identity and leverage time anchors to improve robustness under overlap and rapid turn switching [10]. Overall, these efforts mainly strengthen speaker grounding, namely identifying who spoke when and what, while interaction-level reasoning over speaker relations, motivations, and evolving speaker states remains comparatively underexplored.

Evaluation infrastructure has not kept pace with these modeling advances. Existing speech benchmarks predominantly focus on single-speaker settings or isolated subtasks [11, 12], such as ASR, speaker verification, emotion recognition, and intent detection on individual utterances. Conversation-level evaluations and diarization studies [13, 14] typically measure transcript quality or speaker boundary errors, but provide limited diagnosis of speaker-centric understanding in realistic multi-speaker conversations. Consequently, it remains unclear how current LALMs perform across the full spectrum from fine-grained speaker attribution to higher-order multi-speaker reasoning, and which capabilities constitute the primary bottle-

Table 1: *Data sources used in MSU-Bench. We include conversational and media-style corpora to evaluate robustness across domains.*

Domain	Data Source	Duration
Conversational corpora	Chinese Telephone ¹	5h
	English Telephone ²	5h
	AliMeeting [15]	12h
	CHiME-6 [16]	12h
Media-style corpora	Wild English Podcast	66h
	Wild Chinese Podcast	31h
	Wild Chinese Movie	400h
	Wild English Movie	200h

necks.

To bridge this gap, we introduce MSU-Bench, a speaker-centric benchmark for realistic multi-speaker conversations. MSU-Bench adopts a two-tier hierarchy from speaker grounding to multi-speaker reasoning, covering 16 tasks across five capabilities. To instantiate this task framework, we build a scalable annotation and QA generation pipeline with human-in-the-loop verification, ensuring question validity, answer determinacy, and label correctness. This process results in 2,300 QA instances for objective and diagnostic evaluation. Finally, we analyze multiple speaker-referencing schemes and diagnostic error types, providing actionable insights into the bottlenecks of current LALMs.

2. MSU-Bench: Hierarchical Design for Multi-Speaker Understanding

MSU-Bench evaluates speaker-centric understanding in realistic multi-speaker conversations through a two-tier task hierarchy and diagnostic multiple-choice QA. The benchmark instances are constructed using a scalable annotation and QA generation pipeline with human-in-the-loop verification.

2.1. Benchmark Design and Task Hierarchy

MSU-Bench is designed around five dimensions: multi-tier, multi-speaker, multilingual, multi-scenario, and multi-task. It progresses from speaker grounding to multi-speaker reasoning, requires at least two speakers per instance, covers Chinese and English, and includes 16 speaker-centric tasks across five capabilities. To evaluate robustness across domains, we sample from eight corpora spanning telephone conversations, meetings, podcasts, and movies, as summarized in Table 1.

The task suite follows a two-tier diagnostic hierarchy. Figure 1 illustrates this hierarchy through representative speaker anchors and QA examples, while Table 2 details the corresponding capabilities, definitions, and representative tasks. Tier 1 focuses on speaker-centric identification, requiring models to ground attributes, identities, and speaker-specific information to particular speakers. Tier 2 targets multi-speaker reasoning over context and discourse structure, including speaker relations, dialogue roles, motivations, and interaction dynamics.

To vary speaker grounding difficulty, we define five speaker-referencing schemes. No index uses a target-speaker audio snippet as direct acoustic grounding. Time index refers to the target speaker through a specified time span, whereas

¹Mandarin Chinese Conversational Speech Corpus – Telephony, a publicly available corpus from MagicHub.

²English Conversational Speech Corpus – Telephony, a publicly available corpus from MagicHub.

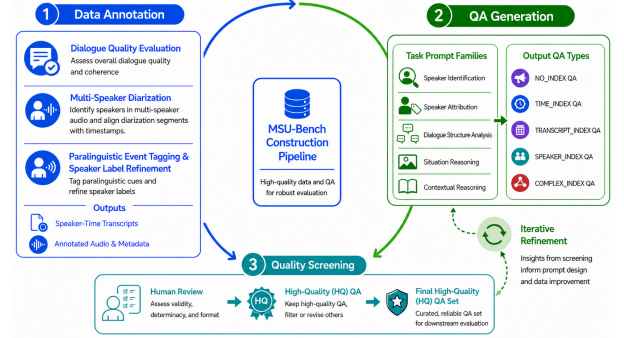


Figure 2: *MSU-Bench construction pipeline. The pipeline consists of dialogue quality assessment, speaker-aware annotation, speaker-referenced QA generation, and human-in-the-loop quality control.*

transcript index identifies the target speaker by a quoted transcript. Speaker index identifies the target speaker according to the speaker’s order of appearance in the dialogue. Complex index combines multiple cues, such as time spans and transcript excerpts, requiring alignment across complementary references.

2.2. Data and QA Construction Pipeline

Table 1 summarizes the public Chinese and English multi-speaker corpora used to construct MSU-Bench, covering both conversational corpora and media-style audio across diverse acoustic conditions. We apply unified preprocessing, including resampling, mono conversion, and channel selection for movie audio. To preserve interaction cues under practical input constraints, audio is segmented into short clips of 1–2 minutes and long clips of 2–5 minutes.

Figure 2 illustrates the QA construction pipeline. First, Gemini performs dialogue quality assessment to select informative and coherent dialogue segments for subsequent annotation. Second, high-quality segments are annotated with multiple types of information: speaker diarization and transcript annotations are produced through the Volcano API, while Gemini is used to annotate speaker identity, sound events, and paralinguistic cues [20, 21]. Third, conditioned on the annotations, raw audio, and task-specific prompts, Gemini generates QA candidates under the predefined speaker-referencing schemes. Finally, trained human annotators verify the metadata, revise invalid or ambiguous questions, check answer determinacy and format compliance, and retain only qualified items. This process produces 2,300 verified QA instances.

2.3. Evaluation Protocol

MSU-Bench uses four-option multiple-choice QA for deterministic scoring. Each question has exactly one ground-truth option supported by explicit audio evidence, and models are required to output a single option letter (A/B/C/D). We use exact-match accuracy as the primary metric and report results by task, capability group, tier, and speaker-referencing scheme.

The three incorrect options are designed for diagnostic error analysis: wrong-speaker options use information from the dialogue but attribute it to the wrong speaker; hallucination options introduce plausible but unsupported content; and unknown options test whether a model incorrectly treats an answerable question as indeterminate. Because each distractor is annotated with its intended failure mode, a model’s choice of an incorrect option can be mapped to a diagnostic error type. Instruction-

Table 2: *Hierarchical task taxonomy of MSU-Bench. Tasks progress from speaker grounding to reasoning over multi-speaker context.*

Capability	Definition	Representative Tasks
Tier 1: Speaker Grounding and Identification		
Speaker Identification [17] (SID)	Ground speaker identities and speaker-specific information across dialogue segments.	• Reverse Speaker Retrieval (RSR) • Speaker Retrieval (SR) • Speaker-specific Viewpoint Summarization (SVS) • Speaker Counting (SC) • Speaker Verification (SV) • Accent Identification [18] (AI) • Age Recognition [19] (AR) • Gender Identification (GI) • Emotion Recognition (ER) • Speaker Profiling (SP)
Speaker Attribute Recognition (SAR)	Infer core speaker attributes and a brief profile from multi-speaker conversational audio.	
Tier 2: Multi-Speaker Dialogue Reasoning		
Dialogue Scene Reasoning (DSR)	Infer dialogue background, speaker roles, and situational context from multi-speaker evidence.	• Background Inference (BI) • Role/Identity Identification (RII)
Dialogue Structure Analysis (DSA)	Recognize dialogue acts and question-answer relations across turns.	• Dialogue Act Recognition (DAR) • Q&A Structure ID (QASI)
Dialogue Contextual Reasoning (DCR)	Reason over multi-speaker context for emotion interactions and viewpoint aggregation.	• Emotion Interaction Reasoning (EIR) • Multi-speaker Viewpoint Summarization (MSVS)

following errors are counted separately when a model fails to produce a valid single option letter.

3. Experimental Setup and Results

We evaluate nine speech-language models on MSU-Bench, including six open-source models and three closed-source Gemini systems. The open-source models include Qwen2.5-Omni, Qwen3-Omni [22], AudioFlamingo-3 [23], Kimi-Audio [24], StepAudio2 [25], and MiMoAudio [26], covering both omni-style and audio-oriented architectures. The closed-source models include Gemini-2.5-Flash, Gemini-2.5-Pro, and Gemini-3-Flash, evaluated through their official APIs. All models are tested zero-shot with the same instruction template, which specifies the target task and requires exactly one option letter from A, B, C, and D. We use exact-match accuracy against the verified ground-truth option, without task-specific fine-tuning or few-shot demonstrations. Since Gemini is also involved in QA construction, Gemini-based evaluations may be affected by potential instruction-format preferences. To reduce this effect, all QA items are manually reviewed and revised before inclusion to ensure answer determinacy, format consistency, and model-independent validity.

Table 3 reports exact-match accuracy by tier and task. Overall performance varies substantially across models, ranging from 0.19 for Qwen2.5-Omni to 0.77 for Gemini-3-Flash. Among open-source models, MiMoAudio performs best, achieving 0.56 overall and the strongest open-source averages on both Tier 1 and Tier 2, with 0.52 and 0.64, respectively. StepAudio2 and Kimi-Audio follow with close overall scores of 0.44 and 0.43, while AudioFlamingo-3 and Qwen3-Omni both obtain 0.39 overall. At the task level, performance is not uniform across capabilities: tasks requiring speaker attribution, viewpoint aggregation, and dialogue reasoning remain challenging for most open-source models. In contrast, models tend to perform better on tasks with clearer acoustic or semantic cues, such as background inference and some speaker-attribute recognition tasks. This pattern suggests that MSU-Bench exposes not only overall model differences but also fine-grained

weaknesses across speaker-centric capabilities.

Closed-source systems consistently outperform open-source systems. Gemini-3-Flash achieves the best overall score and the highest averages on both tiers, reaching 0.73 on Tier 1, 0.84 on Tier 2, and 0.77 overall. Gemini-2.5-Pro and Gemini-2.5-Flash also perform strongly, with overall scores of 0.70 and 0.69, respectively, both exceeding the strongest open-source model. These results suggest that recent closed-source systems have made substantial progress on speaker-centric understanding, but the remaining errors analyzed in Section 4 show that complex speaker grounding and interaction reasoning are still not fully solved.

Among open-source models, MiMoAudio remains the strongest system, especially on Tier 2 reasoning tasks. In contrast, the Qwen-Omni systems do not consistently outperform audio-oriented models, suggesting that broad multimodal capability does not necessarily translate into robust speaker-centric understanding.

4. Analysis and Discussion

We further analyze model behavior and benchmark quality from three diagnostic perspectives: speaker grounding under different speaker-referencing schemes, diagnostic error-type composition under objective QA, and human verification of QA quality.

4.1. Speaker-Referencing Scheme Analysis

Table 4 reports model performance under three representative speaker-referencing schemes. No Index provides a target-speaker audio snippet for direct acoustic grounding, while Time Index requires models to locate the target speaker from a specified time span. This makes Time Index the most challenging setting for most models, especially under overlap and rapid turn switching. Complex Index combines time spans, transcript excerpts, and speaker-related cues, offering additional localization information that can help models disambiguate speakers.

The results show that speaker-referencing schemes substan-

Table 3: **Exact-match accuracy on MSU-Bench by tier and task.** We compare speech-language models across Tier 1 (Identification) and Tier 2 (Understanding). Best results in each column are highlighted in **bold**, and second-best results are underlined. Avg columns report instance-level average accuracy.

Models	Tier 1 (Identification)										Tier 2 (Understanding)								Avg
	AI	AR	GI	ER	SP	RSR	SR	SVS	SC	SV	Avg	EIR	MSVS	BI	RII	DAR	QASI	Avg	
Qwen2.5-Omni [27]	0.19	0.20	0.24	0.16	0.21	0.09	0.19	0.24	0.10	0.24	0.19	0.24	0.39	0.00	0.21	0.24	0.16	0.21	0.19
AudioFlamingo-3[23]	0.43	0.39	0.41	0.30	0.35	0.55	0.23	0.60	0.43	0.29	0.40	0.38	0.32	0.69	0.24	0.31	0.36	0.38	0.39
Qwen3-Omni[22]	0.30	0.35	0.66	0.20	0.48	0.21	0.49	0.47	0.45	0.40	0.40	0.18	0.52	0.78	0.28	0.29	0.23	0.38	0.39
Kimi-Audio[24]	0.47	0.37	0.41	0.37	0.37	0.24	0.46	<u>0.68</u>	0.51	0.24	0.41	0.44	0.38	0.75	0.41	0.42	0.42	0.47	0.43
StepAudio2[25]	0.44	0.46	0.51	0.38	0.52	0.24	0.34	0.58	0.48	0.43	0.44	0.52	0.51	0.44	0.39	0.48	0.39	0.46	0.44
MiMoAudio[26]	<u>0.59</u>	0.42	0.58	0.49	0.52	0.45	0.38	0.68	0.58	0.48	<u>0.52</u>	0.58	0.57	0.72	0.68	0.72	0.57	0.64	0.56
Gemini-2.5-Flash	0.49	0.55	0.75	<u>0.54</u>	0.71	0.73	0.66	<u>0.72</u>	0.55	<u>0.68</u>	<u>0.64</u>	<u>0.71</u>	0.78	<u>0.81</u>	<u>0.71</u>	0.79	<u>0.83</u>	<u>0.77</u>	0.69
Gemini-2.5-Pro	0.56	<u>0.62</u>	0.83	0.52	<u>0.71</u>	<u>0.79</u>	<u>0.67</u>	0.69	<u>0.67</u>	0.68	<u>0.67</u>	0.67	<u>0.82</u>	0.69	0.71	<u>0.80</u>	0.77	0.74	0.70
Gemini-3-Flash	0.69	0.65	<u>0.83</u>	0.58	0.70	0.88	0.69	0.84	0.70	0.75	0.73	0.84	0.85	0.84	0.79	0.83	0.91	0.84	0.77

Model	Tier 1 Accuracy			Tier 2 Accuracy		
	No	Time	Cpx	No	Time	Cpx
Qwen3-Omni	<u>0.57</u>	0.38	0.46	0.34	0.28	0.35
MiMoAudio	0.53	<u>0.54</u>	<u>0.60</u>	<u>0.64</u>	<u>0.53</u>	<u>0.63</u>
Gemini-3-Flash	0.71	0.64	0.84	0.83	0.76	0.92

Table 4: **Accuracy under representative speaker-referencing schemes.** No Index, Time Index, and Complex Index represent direct acoustic grounding, temporal grounding, and combined-cue grounding, respectively.

Model	Tier 1 Error Type Rate				Tier 2 Error Type Rate			
	WS	HAL	UNK	INS	WS	HAL	UNK	INS
Qwen3-Omni	0.14	0.05	0.27	0.00	0.18	0.08	0.40	0.00
MiMoAudio	0.28	0.08	0.08	0.16	0.53	0.11	0.09	0.13
Gemini-3-Flash	0.30	0.07	0.05	0.13	0.67	0.11	0.03	0.06

Table 5: **Diagnostic error-type composition.** WS, HAL, UNK, and INS denote wrong speaker, hallucination, unknown, and instruction-following failure, respectively.

tially affect model accuracy. Time Index generally yields the lowest accuracy, indicating that temporal grounding remains a major bottleneck in multi-speaker dialogue understanding. In contrast, Complex Index often improves performance by providing complementary grounding cues. Gemini-3-Flash performs best across all schemes, achieving 0.71, 0.64, and 0.84 on Tier 1, and 0.83, 0.76, and 0.92 on Tier 2 under No Index, Time Index, and Complex Index, respectively. Among open-source models, MiMoAudio is more robust under Time Index and Complex Index. Overall, additional grounding cues improve speaker-centric QA, while time-based localization remains a key weakness.

4.2. Diagnostic Error-Type Analysis

Table 5 summarizes the diagnostic error-type composition for representative models. Qwen3-Omni shows the highest unknown-error rate, increasing from 0.27 in Tier 1 to 0.40 in Tier 2. This suggests that when a model lacks sufficient multi-speaker grounding or dialogue reasoning ability, it tends to choose the unknown option rather than make an unsupported prediction.

For stronger models, errors shift mainly toward wrong-speaker choices. MiMoAudio reaches a Tier 2 wrong-speaker

Metric	Tier 1	Tier 2
Initial QA validity (human-judged)	95%	86%
Human-GT answer agreement	98%	96%

Table 6: **Human verification results.**

rate of 0.53, while Gemini-3-Flash further rises to 0.67. This indicates that models with basic multi-speaker QA capability are less limited by answer indeterminacy and more affected by fine-grained speaker attribution errors. Hallucination and instruction-following failures remain secondary overall. Thus, as task performance improves, the dominant bottleneck shifts from unknown responses to assigning evidence to the correct speaker in complex interactions.

4.3. Question Quality and Human Verification

Open-ended QA better reflects real-world LALM interactions, but free-form outputs make evaluation noisy and costly. They require answer normalization and often an additional LLM judge, which can introduce extra errors. We therefore evaluate MSU-Bench using objective single-choice questions and involve eight researchers with audio backgrounds to verify and revise all QA instances for quality assurance. During verification, the initial QA candidates reach human-judged validity of 95% in Tier 1 and 86% in Tier 2; invalid or ambiguous items are revised or removed before inclusion. We further ask the same annotators to answer the retained questions to verify the final labels, achieving 98% agreement with the verified ground truth in Tier 1 and 96% in Tier 2, as shown in Table 6. These results indicate that our QA protocol provides reliable evaluation labels for speaker-centric assessment.

5. Conclusion

We presented MSU-Bench, a speaker-centric benchmark for realistic multi-speaker conversations with a two-tier hierarchy, 16 tasks, and 2,300 verified QA instances. Through evaluations of nine speech-language models, we show that speaker-referencing schemes and diagnostic error types reveal persistent bottlenecks: temporal grounding is especially difficult, strong models still suffer from wrong-speaker attribution, and weaker models often default to unknown under higher reasoning demands. These findings provide a diagnostic basis for improving robust multi-speaker audio understanding.

6. Acknowledgements

This research is supported by National Natural Science Foundation of China (Grant No. 62401377).

7. Generative AI Use Disclosure

Generative AI tools were used in two distinct capacities in this work. As part of the research methodology, Gemini was employed in the MSU-Bench construction pipeline for dialogue quality assessment, paralinguistic annotation, and QA generation (detailed in Section 2.2). All AI-generated annotations and QA items were subject to rigorous human-in-the-loop review by trained annotators, who verified metadata correctness, question validity, answer determinacy, and answer-format compliance before any item was included in the benchmark. For manuscript preparation, AI writing assistants were used to improve clarity and grammar in certain passages of the text.

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